

WHITE COMFORT EXECUTIVE TRANSPORT LTD

Engineering Services & IP Assignment Agreement

This Agreement is made and entered into this **16th day of December, 2025**, by and between:

White Comfort Executive Transport Ltd, a company incorporated under the laws of Ghana, with its principal office at Sofoline, KOAN Filing Station (hereinafter referred to as “**Company**”),

AND

The undersigned engineers:

1. Parwar William Ofosu, **Engineering Lead / Backend Engineer (NodeJS/ExpressJS/NestJS)**
2. Jerry John Rockson, **Mobile Engineer (React Native)**
3. Lemuel O. Abrokwah, **Frontend Engineer (Reactjs)**
4. **Daniel Obeng, UI/UX Designer (Figma)**

(Collectively referred to as “**Engineers**”)

1. Scope of Work

The Engineers shall design, develop, test, and deploy the **Comfort App**, an intracity and intercity transport mobile application, including but not limited to:

Product Components

1. **Passenger App (Android/iOS) – React Native**
2. **Driver/Conductor App (Android/iOS) - React Native**
3. **Admin Console (Web Dashboard) - ReactJS**
4. **Backend APIs + Cloud Infrastructure - ExpressJS**

1.2 Passenger Application — Feature Scope

Core Features (MVP)

MVP Goal: Support 1,000 daily API requests, 10,000 concurrent users

1. Live Tracking & Real-Time ETA:

- Live and interactive map view showing moving icons of active trotros on their routes.
- A list view of nearby trotros, sorted by estimated time of arrival (ETA) to the user's selected designated stop.
- Clear visual markers for all official "COMFORT" designated bus stops.

2. Route Selection (Intra-city + Inter-city)

3. Mobile Money Payments (MoMo)

- Secure integration with a major Mobile Money API (e.g., MTN MoMo, Vodafone Cash).
- Auto fare computation: A seamless flow for passengers to pay fares directly through the app before or at the end of a trip.
- Digital receipts
- Wallet & payment history

4. Booking System (Inter-city only)

- Reserve seats
- Book specific departure times
- Ticket QR code verification

5. User Account & Preferences

- OTP phone login
- User profile
- Saved routes
- Trip history

1.3 Driver/Conductor Companion Application (MVP Feature Set)

Core Features

1. This application is critical for providing the data that powers the platform.
 - a. Automatic Fare Calculation & Management:
 - i. Fare is calculated dynamically based on the exact distance traveled using GPS data.
 - ii. The pricing algorithm (e.g., rate per km) is controlled remotely by the Admin panel.
 - iii. The app displays the calculated fare for the passenger.
 - b. Mobile Money Payment Acceptance:
 - i. A simple interface to view, confirm, and manage incoming payments from passengers via the app.
 - c. 3. Route & Availability Management:
 2. A "Start Route"/"End Route" button to begin and conclude a journey, which triggers live location sharing.

Unified App Flow

Example: Kumasi → Adenta (Accra), Kumasi → Santasi (intra-city)

No stations. No terminals. Pure on-demand stop-to-stop mobility.

- Key:
 - Passengers should move entirely without fixed stations or terminals
 - Vehicles operate dynamically and pick up from validated points
 - Same user flow for intra-city & inter-city trips
 - The app decides the category (short trip vs long trip) invisibly

This removes friction and keeps the user experience **simple, unified, and scalable.**

1. Home Screen – City + Destination Search

When the app opens:

User inputs only TWO things:

1. **Pickup Location** → GPS auto-location or manual search
2. **Destination** → Address, community, town, region

The system automatically detects:

- Whether the trip is **intra-city, inter-city(inter-regional)**
- Available COMFORT vehicles on that route
- Estimated time for nearest vehicles

NO station selection. NO terminals.

Just Enter A → B.

2. Smart Trip Classification (Behind the Scenes)

The backend classifies the trip without showing categories:

Trip Type	Example	Internal Logic
Intra-city	Ahodwo → Santasi (Kumasi)	< 25 km inside same city
City-to-City (Inter-city)	Kumasi → Adenta (Accra)	> 25 km & cities differ

This ensures **user experience remains identical** across Ghana.

3. Vehicle Discovery (Live Map)

The app shows:

- Nearby active COMFORT vehicles
- Available seats
- Expected Arrival Time
- Trip estimate (distance + time)

For longer routes, some options may show:

- “Direct Trip”
- “One Transfer Required” (if no direct vehicle available)

4. Seamless Pickup Experience

Once a vehicle is selected:

- App suggests **closest safe pickup point** (algorithm-generated)
- Examples:
 - Roadside landmark
 - Nearby junction
 - Bus stop (optional)
 - Popular local points

Driver/Conductor sees the pickup location with map navigation.

Passenger sees:

- Driver name
- Vehicle details
- Real-time movement toward pickup point

5. Trip Start – Unified Logic for All Trip Types

Once the vehicle arrives:

- Passenger boards
- Driver taps **Start Trip**
- GPS distance tracking begins
- Fare accumulates based on:
 - Distance traveled
 - Pricing rule for the trip category

Exactly the same process for intra-city and inter-city.

6. In-Trip Experience

Passenger views:

- Current route on map
- Remaining distance/time
- Number of stops ahead
- Seat occupancy
- Safety features (SOS button)

Driver App:

- Auto-updating distance
- Seat count
- Next pickup/drop-offs (if multiple passengers)

7. Seamless Drop-off (No Terminal Required)

Passenger chooses:

- “Drop me here”
- Or selects from automatic suggested drop-off points (safe roadside spots)

Driver ends trip → App locks the fare.

8. Payment Process

Payment is automatic and unified:

- MTN MoMo

No cash unless the driver chooses hybrid mode.

Fare breakdown shown:

- Distance
- Trip category (calculated internally)
- Service fees
- Safety surcharges (optional)

The passenger gets a digital receipt.

9. Post-Trip Actions

Passenger:

- Rates driver
- Saves frequent routes
- Views trip summary
- Option to rebook same trip

Driver:

- Daily revenue dashboard
- Trip logs
- Heatmap for high-demand areas

End-to-End Flow Summary

Search → See Vehicles → Select → Pickup → Trip → Drop-off → Pay → Rate

No stations, no route categories, no complexity.

1.4 Admin Panel

A secure web-based dashboard for system oversight and management.

1. Dynamic Pricing Engine: Interface to set, adjust, and manage the base fare rates (e.g., GHS per kilometer) that power the automatic fare calculation in the Driver app.

2. Driver & Vehicle Management: Tools to onboard, verify, and manage company employed drivers and their vehicle information.

3. **Financial Dashboard:** A view of all transactions processed through the platform, including total revenue, trip count, and payment success rates.
4. **System Monitoring:** View system alerts, app health status, and live active routes.
5. **KPI Reporting:** Basic reporting on the key success metrics (user growth, trips per day, etc.).

PROJECT SCOPE (ACTIVITIES & DELIVERABLES)

In Scope

- Mobile apps for passengers + drivers
- Web admin dashboard
- Backend APIs
- Mapping + MoMo integrations
- Deployment to Play Store, App Store
- Infrastructure setup (hosting, CI/CD)
- UX research + UI/UX design
- Unit testing + UAT
- Documentation (API docs, user manuals, SOPs)

Out of Scope (Phase 2)

- Web version of the Passenger App (Progressive Web App - PWA).
- Loyalty points system
- Advanced features like route planning, multi-modal transport integration, or loyalty programs.
- Carpooling/rideshare features

Offline booking centers integration

The work shall be completed according to the following **3-month timeline**:

Month Deliverables

- 1 UI/UX Design, Core Development
- 2 MVP Completion, Internal Testing
- 3 User Testing, Deployment

Payments will be made within 5 business days of approval of each milestone by the Company.

2. Intellectual Property

- All **source code, designs, documentation, databases, and other materials** created by the Engineers under this Agreement shall be the **exclusive property of White Comfort Executive Transport Ltd.**
- Engineers shall **not retain any rights, copies, or reuse** of the IP without written consent.
- Engineers assign to the Company all rights, title, and interest in the Comfort App.

3. Confidentiality & NDA

- Engineers shall not disclose, publish, or use **any Company or user information** for any purpose other than fulfilling this Agreement.
- This obligation applies **during and after the term** of this Agreement.
- **Immediate termination** applies for breach of confidentiality.

4. Data Privacy

- All **user, driver, vehicle, trip, and payment data** belong exclusively to the Company.

- Engineers shall **not retain any copies** of such data post-termination.

5. Termination

- **Immediate termination** for breach of this Agreement, including confidentiality or IP violations.
- **Non-breach termination** requires **7 days written notice**.
- Upon termination, Engineers must return all Company property and data.

6. Representations & Warranties

Engineers represent that:

- They have the skill and expertise to perform the work.
- Work delivered will be original and not infringe third-party rights.

7. Liability & Indemnity

- Engineers shall indemnify and hold the Company harmless from any claims arising from intellectual property infringement or negligence.

8. Dispute Resolution

- Parties shall attempt **mediation** in case of disputes.
- If unresolved, disputes shall be governed by **Ghanaian law** and subject to the jurisdiction of Ghanaian courts.

9. Force Majeure

Neither party shall be liable for failure or delay due to events beyond their reasonable control, including natural disasters, strikes, or government action.

10. Entire Agreement

This Agreement constitutes the **entire agreement** between the parties and supersedes all prior understandings, whether written or oral.

11. Severability

If any provision of this Agreement is held invalid, the remaining provisions shall remain in full force and effect.

13. Post-Deployment Support

- Any additional maintenance or bug fixes after the 3-month deployment period will be covered under a **separate paid maintenance contract**.

Signatures

For White Comfort Executive Transport Ltd

Name: _____

Title: _____

Date: _____

Engineers:

1. Name: Parwar William Ofosu Signature: Parwar Date: 17/12/2025

2. Name: Daniel Kobby Obeng Signature: DKO Date: 17/12/2025

3. Name: _____ Signature: _____ Date: _____

4. Name: _____ Signature: _____ Date: _____

